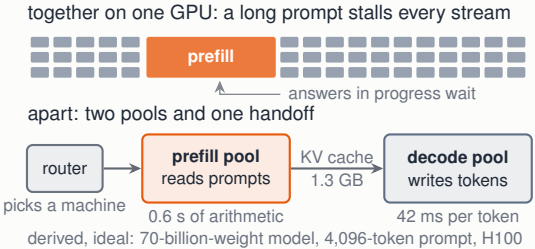
Disaggregation: give each phase its own machines

Reading the prompt and writing the answer stress different parts of a chip, so run them on separate pools.



INTUITION A kitchen with a prep station and a plating line. A big chopping job no longer blocks the plates going out.

Why split the two phases

Two different jobs. Prefill reads the whole prompt in one pass and is limited by arithmetic speed. Decode writes one token per pass and is limited by how fast memory can be read.

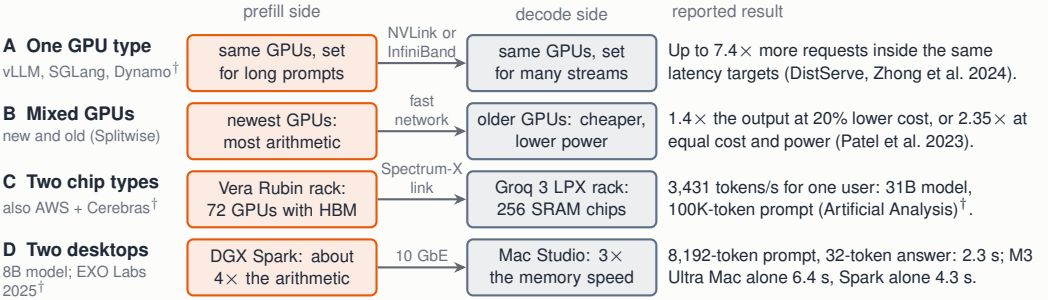
Together they collide. A long prompt arriving on a shared GPU delays every answer already streaming.

Apart, each pool is sized and tuned alone. DistServe served up to 7.4× more requests inside the same latency targets (Zhong et al. 2024).

NEW TERMS **Disaggregation**: splitting one job across separate groups of machines (**pools**), each suited to its part. **KV (key-value) cache**: the saved result of prefill, reused by every decode step.

Four ways to build it

The same split, from identical GPUs in two roles to two different kinds of chip, and down to two desktops.



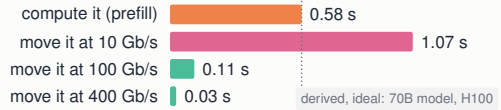
INTUITION At every scale: the best calculator reads the prompt, the fastest memory writes the answer.

NEW TERMS **HBM**: high-bandwidth memory beside a datacenter GPU. **SRAM**: memory inside the processor chip itself: fastest, but small. **NVLink**, **InfiniBand**: fast datacenter links. **10 GbE**: 10-gigabit Ethernet.

Where the split pays, and where it does not

It pays when prefill is a large share of the work and the cache can be moved faster than it is computed.

one 4,096-token prompt: compute its cache, or move it?



The cache is 1.3 GB. Break-even is a link of about 18 Gb/s. Above it, the cache can stream out layer by layer during prefill, so the handoff adds almost nothing. A slower prefill machine tolerates a slower link: pair D works at 10 Gb/s.

INTUITION Sending work to a specialist pays only if the handoff is quick next to the job: across the hall, yes; by mail for a five-minute task, no.

Worth it when (after Mitra et al. 2025)

+ prompts are long and answers short
+ the model is large and runs on many GPUs
+ first-token and streaming targets are both tight
+ the two sides differ: compute-rich, memory-fast

Not worth it when

– there is one machine, or only a few GPUs
– prompts are short, answers long, or the model small
– most of the prompt is already cached (chat, agents)
– the link is slower than the prefill itself

NEW TERMS **Gb/s**: gigabits per second; 8 Gb/s moves 1 GB each second. **Rate matching**: re-sizing the two pools as traffic shifts (Mitra et al. 2025). **Prefix cache**: a KV cache kept for text many requests share.